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Table of Contents

1. INTRODUCTION ..... **Error! Bookmark not defined.**
2. PURPOSE 5
3. DEFINITIONS 5
4. OBJECTIVE 5
5. RESPONSIBILITIES 6
6. REFERENCES 6
7. COST AND BUDGET 7
8. SCHEDULE 7
   8.1 Work Breakdown Structure (WBS) 7
   8.2 Project Schedules 7
   8.3 Original Master Schedule 8
   8.4 Original Work Schedule - Lower Level Schedule 8
   8.5 Overall Project S-Curve 8
   8.6 Lower Level S-Curve 8
   8.7 Updating,Tracking Schedules, S-Curves and Recovery Plan 9
9. REPORTING 10
   9.1 General 10
   9.2 Weekly Report 10
   9.3 Monthly Report 11
   9.4 Close Out Report 12
10. PROGRESS MEASUREMENT 12
   10.1 General 12
   10.2 Actual Progress 12
   10.3 Progress Measurement Criteria and Weight Factors 13
11. PHYSICAL PROGRESS MEASUREMENT 13

1 INTRODUCTION

The Forel field is an offshore oil field prospects located in Block B of the Natuna Sea, 24 km of Northwest Belanak and has 300 feet water depth in average. Forel field was discovered in 1984 by Forel-1A well exploration drilling and followed by the Forel-2 well in 1985, while the Bronang field is a gas-producing field located 40 km to the northwest of Belanak Field or 70 km to the Southwest form the Hang Tuah MoGPU. Bronang field was discovered in 1993 by Bronang-1 well exploration drilling Forel field development require gas injection source for pressure maintenance and lifting gas to the reservoir. Gas lift source is also required to optimize Forel's crude oil production. The requirement of gas injection and gas lift at Forel will be supplied by Bronang field.

The design of Forel and Bronang field production facilities are dry tree concept where all production weels will be placed on a wellhead platform (WHP) that is designed unmanned with the concept of the Minimum Facility Platfrom (MFP) which is designed with a simple technology system but has a high level of reliability and requires minimum intervention and maintenance.

Forel MFP will be routed to Floating Production, Storage and Offloading (FPSO) via multiphase flexible pipeline for further processing to meet crude oil market requirement, while the gas is boosted to meet the gas lift and gas injection pressure requirement. The Bronang production will be also routed to FPSO via Forel WHP to supply gas injection at initial production phase and fuel gas throughout Forel production lifetime.

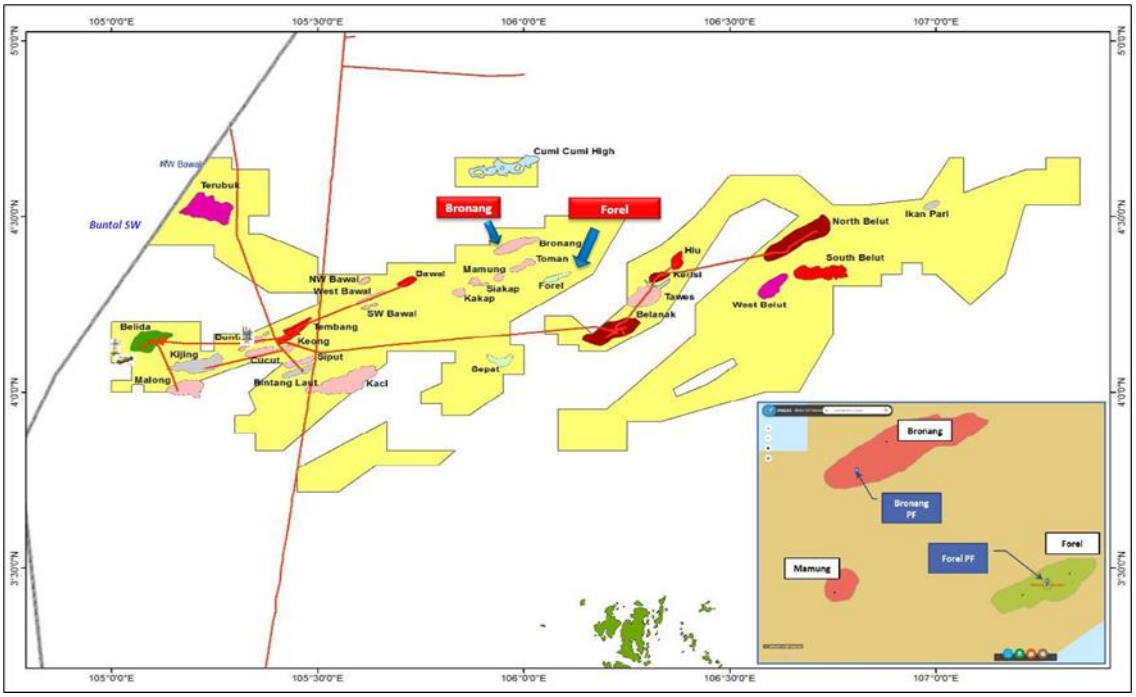

Figure 1 : South Natuna Block B PSC Location Map

2. PURPOSE

This procedure establishes the requirements that Project must meet in Planning, Scheduling, Monitoring and Evaluating projects managed by the OWNER in accordance with the project Scope of Work.

The purpose of this procedure is to lay down guidelines for efficient execution of project construction from order stage to the completion of project handing over according to contractual obligations within the stipulated cost and time.

3. DEFINITIONS

"**CHARTERER**" means Medco E&P (Natuna) Ltd.

"**OWNER**" means Consortium Haniochem – Cakra - MTCE

4. OBJECTIVE

Project Control aimed is achieving the project objectives in the area of cost and schedule, that is, to complete the project within the defined project completion date and budget.

The objectives of Project Control System are to:

- Define realistic work and expenditure plans for executing the project
- Provide summary and exception reporting against these plans to determine corrective actions.

Project Control System integrates the elements of:

- Planning system that involves tasks for the establishment of control baselines (WBS; Weight Factor; Measurement System; Scheduling System; Budgeting System; Planning S-curve, etc.).
- Monitoring and Recording system that involves tasks for gathering data, compiling and updating data and reporting information.
- Analysis and Forecasting system that involves tasks for measuring the difference between the plan and the actual and for predicting the future performance.
- Reporting system that involves tasks for sorting the information in various control levels needed for taking necessary corrective actions.

All of these elements are essential to effective project control.

In Project Control practices, basic control philosophy used is to compare 'the actual against the plans'. Which means that there must be some plans to be used as a basis for comparison (i.e. budget or schedule), and a recording system of actual cost, schedule and progress performance, as well as an adequate measurement system.

5. RESPONSIBILITIES

OWNER Project Control Policy is basically the management directives for the Project Team and other concerned functional group in executing effective project control. This policy also bridges the basic control philosophy with the development of Project Control Procedures for a specific project.

The following is OWNER Project Control Policy Statement:

1. OWNER Project Control Systems will place the emphasis on *early warning capabilities* to spot deviations between the actual cost and schedule performances and their respective plans. Thus, it is the Project Control Team responsibility to alert Project Manager to any condition, which might impact the final cost and/or completion date of the project.

2. Project control activities should be *coordinated* under the Project Control Manager that is directly responsible to the Project Manager. Project Control Manager heads the Project Control Team comprising the Planning, Scheduling and Cost controlling groups which are responsible for the coordination of cost and schedule control activities respectively.

3. The principal responsibilities of Project Control Team in the implementation of project control are as follows:
   - to plan and provide the Project Manager with the means of controlling cost and schedule, at the required level of detail, for the effective execution of the project.
   - to manage and implement the Project Control Systems throughout the life of the project.
   - to assess, analyse and report the impact of all deviations from project budget and schedule, at the earliest possible time, for evaluation and remedial actions by the Project Manager

6. REFERENCES

CONTRACT document "General Terms and Condition"
CONTRACT Exhibit B "Scope of Work"
CONTRACT Exhibit C "Compensation"
CONTRACT Exhibit F " Administration and General Requirements"

7. COST AND BUDGET

Upon contract award, price proposal becomes the contract price. And OWNER Project Control Department establishes the Budget based on such contract price.

As further information is developed, the Project Control Manager establishes the Preliminary Budget, which is basically developed from the Budget Estimate. And final budget shall be approved by OWNER Project Manager and Project Management.

The Cost engineer will be responsible for all cost control and reporting.

A Cost Control program will be developed and utilised for Cost Commitment Tracking with approved budget. This will be used to track, record and report, project commitments, expended costs and approved variances on the project.

Costs will be monitored as work progresses to ensure that all costs remain within the Contract Price budget and corrective actions will be taken when the forecast of any cost component begins to trend vary than the cost estimate.

OWNER will generate all the cost report based on Project Work Break Down Structure (WBS) and OWNER shall define Cost Breakdown Structure (CBS) and its codification for proper monitoring for project cost under each heads.

8. SCHEDULE

8.1 Work Breakdown Structure (WBS)

A WBS defines the Work to be accomplished and the way in which the Work is to be performed.
The objective is to subdivide the total project into manageable units of work for effective planning and control. The Work is subdivided into successive levels of detail that consist of elements that identify the different areas and functional requirements of the Work. Each lower level is a meaningful subdivision of a higher level element. Each unit of the WBS should have a unique number and a weight - percentage. The WBS will be provided as part of the Original Master Schedule described below and be used as the basis for performance monitoring and reporting, cost and schedule integration.

8.2 Project Schedules

Project Schedule, as defined in CONTRACT Document Exhibit H, "Time Charter Schedule", of this Contract, consists of all of the various schedules at different levels of detail as discussed herein.

8.3 Original Master Schedule

Immediately after Contract Effective Date, but before the project kick off meeting, OWNER will submit a Level-3 Schedule of the Work for CHARTERER approval. The schedule will be in the form of a Gantt Chart and a corresponding CPM Network (Arrow Diagram or Precedence Diagram). The schedule will clearly indicate all major project activities, WBS number, duration, early start and early finish dates, latest start and latest finish dates, critical activities and the critical path including period of performance as described in Exhibit H. The major groups of activities must correlate to items in the Breakdown of the Lump Sum Price and the WBS. The schedule must be sufficiently detailed to allow for reasonable control of all critical path activities. Once the schedule is approved by CHARTERER, the schedule shall be used as an Original Master Schedule of the Work and the base line for S-Curves.

8.4 Original Work Schedule - Lower Level Schedule

One week after CHARTERER approval of the Original Master Schedule, OWNER will submit to CHARTERER a Level-4 CPM Schedule containing engineering and offshore activities. The Level-4 CPM Schedules shall be in the form of a Gantt Chart and Network Diagram and shall clearly indicate all Level-4 activities, WBS number, duration, early start and early finish dates, latest start, latest finish, critical activities and the critical path. Once these schedules are approved by CHARTERER, they shall be used as Original Work Schedules or base lines for the project.

8.5 Overall Project S-Curve

Two week after CHARTERER's approval of the Original Master Schedule, OWNER will submit to CHARTERER an overall Project S-Curve which is based on the weighted WBS and the Original Master Schedule. This overall Project S-Curve shall consist of one curve, based on start dates of the Original Master Schedule. Once this Project S-Curve is approved by CHARTERER, it shall be used as the baseline Project S-Curve.

8.6 Lower Level S-Curve

OWNER will submit lower level S-Curves for engineering, procurement and construction/offshore installation activities (including transportation and mob/ demob). All of these S-Curves shall be made of one curve, shall be based on the start dates of the Original Work Schedule. Once these S-Curves are approved by CHARTERER, these curves shall be used as baselines for working level S-Curves.

8.7 Updating, Tracking Schedules, S-Curves and Recovery Plan

During the project period, OWNER will update the Level-3 and the Level-4 Schedules to reflect the actual dates and current forecasted dates. Each updated schedule must show the original planned dates, the actual dates and the current forecasted dates of each activity. Any planned duration change to the critical items shall be approved by CHARTERER. These schedules shall be updated and submitted to CHARTERER monthly or biweekly or weekly depending upon CHARTERER's request.

During the project period, OWNER will update the overall Project S-Curve and lower level S-Curves to reflect the actual progress of the Work. Each updated S-Curve must show the approved baseline S-Curves. These S-Curves shall be updated and submitted to CHARTERER monthly or biweekly or weekly depending upon CHARTERER's request.

When the overall progress is lagging more than 7% from the current approved planned figure (e.g. actual 20% versus plan 28%) or individual progress of engineering, procurement or fabrication/construction/Conversion/pre-commissioning is lagging more than 10% from the current approved planned figure (e.g. actual 30% versus plan 41%) or a slippage of activities on the critical path of more than 1 (one) week from the current approved schedule during the first 90% of the project duration or any slippage of activities on the critical path during the last 10% of the project duration and without waiting for Charterer to issue any Notice of Schedule Delay, the Ownershall deliver to Charterer a Schedule Recovery Plan. Such Schedule Recovery Plan shall include CPM schedule, S-curves, manpower histograms and a narrative action plan on how it will realistically recover the slippage in accordance with Section 19.2 of the Contract Signature Document. These provisions are to be considered supplementary to the provisions included in Section 19.0 of the Contract Signature Document.

In the event that Owner productivity falls below 95%, Owner shall inform Charterer immediately of such occurrence together with steps taken to remedy said lack of productivity.

9. REPORTING

9.1 General

OWNER will submit weekly and monthly reports periodically as required to CHARTERER including all necessary attachments in hard copies and electronic format.

9.2 Weekly Report

OWNER will prepare and submit a weekly report covering all activities relevant to the previous week's Work. Reports will be updated on a weekly basis and submitted to CHARTERER within one (1) day prior to Weekly Meetings to be held on hard copies and electronic format. Weekly cut-off dates shall be on every close of business Friday or with Mutual understanding with CHARTERER.

The content of Weekly Report, at minimum, shall cover the following items:

- Cover sheet which includes "Week Ending Period"
- Overall Progress Summary
- Area of Concern
- Work Performance
  - Activity This Period and Next Period
    - General
    - Engineering
    - Procurement
    - Logistics
    - Project Control
    - QHSE
    - Construction & Installation
- Attachments
    - Project Progress S-Curve
    - Master Document Register
    - Subcontract Status Report
    - Technical Query Register
    - Updated Project Master Schedule
    - Variation Order Register

In addition to above, CHARTERER may require and OWNER will provide any other additional information as deemed necessary related to weekly reports.

9.3 Monthly Report

OWNER will prepare and submit a monthly report covering all activities relevant to the previous month's Work and such reports shall be updated on a monthly basis. OWNER will submit a draft of first monthly report for CHARTERER review prior to issuance of final monthly report. Monthly cut-off dates shall be on last close of Sunday on each month.

The content of monthly report, at minimum, shall cover the following items:

- Cover sheet which includes "Monthly Period"
- Executive summary
- Area of Concerned
- HSE Summary Report
- Key Date Project
- Overall Progress Summary
- Work Performance (This period and next Period)
  - General
  - Engineering
  - Procurement
  - Logistics
  - Project Control
  - QHSE
  - Construction and Installation
- Commercial
- Attachments
  - HSE Monthly Report
  - Project Progress Curve
  - Master Document Register (MDR)
  - Technical Query Register
  - Procurement Status Report
  - Material Tracking Status
  - Subcontract Status Report
  - Project Master Schedule updated
  - Change Order Register

In addition to this, CHARTERER may require and OWNER shall provide any other additional information as deemed necessary related to Monthly Reports.

9.4 Close Out Report

CONTRACOR will prepare and submit a close out report for CHARTERER review and approval as part of Final Documentation. Such close out report shall be a comprehensive summary of the Work progress achieved from Contract Effective Date to Contract completion.

OWNER will develop the report to cover, at minimum, all sections detailed in the close out report.

OWNER will submit a close out report minimum 1 original and 2 copies complete with soft copy backup in CD-Rom.

10. PROGRESS MEASUREMENT

10.1 General

Actual progress of Work completed shall, at all times, be calculated/measured by OWNER at lowest detail of deliverables based on progress measurement criteria, aggregated to higher levels using associated weight factors, and organized by Work Breakdown Structure.

10.2 Actual Progress

In order to present actual progress of Work completed, OWNER will develop and finalize progress calculation sheets. Progress calculations sheets shall reflect the lowest detail of items/deliverables to be measured and shall consistently tie back to individual activities define in a level-4 Project Master Schedule. Items listed on the progress calculation sheets shall cover all relevant Work defined in Exhibit B “Scope of Work” of Contract, and be structured in accordance with Work Breakdown Structure. Such items shall include, but not be limited to, the following:

- Document deliverables defined in Master Document Register (MDR).
- Procurement for major and bulk materials that shall be tracked from requisition issuance through arrival of materials on site.
- Fabrication/construction/installation which physical quantities fabricated /installed shall be measured and reported.
- Actual progress reflected in % complete figures on each item in such progress calculation sheets, shall be based on deliverable achievements defined under “Progress Measurement Criteria” below. Any Change Orders issued, as necessary, shall be clearly identified and measured as separate line items.

- In addition to this, such deliverables/items listed on progress calculation sheets shall reflect sufficient details for CHARTERER's to justify correctness/accuracy % figures of Work completed.
- Progress calculation sheets, listing level-4 items, including weight factors and approved progress measurement criteria and weight factors, shall be submitted for review and approval, and shall be updated on a weekly basis.
- OWNER shall submit progress calculation sheets to CHARTERER on weekly basis, unless instructed otherwise by CHARTERER.

10.3 Progress Measurement Criteria and Weight Factors

Progress measurement criteria/methods and scales will be developed and submitted for review / approval and approval as separate document **"Project Planning & Progress Measurement Procedure"** and shall be utilized as a basis to define actual progress of Work completed on the above progress calculation sheets. Agreed progress measurement criteria shall not be modified during the course of the Contract execution without approval.

11. PHYSICAL PROGRESS MEASUREMENT

The physical progress of the work shall be determined by measuring the progress of each work unit of CONTRACT.

Progress of lump sum items shall be measured in percentage and for the items of measurable units; progress shall be measured in corresponding units. The progress of these activities shall be recorded as per the break up given in section 8 of this procedure. Progress measurement of any activity, additional to CONTRACT work break down, shall be recorded under 'Change Orders'.

Weightings assigned to each activity is based on CONTRACT price breakup / cost. Progress of all the activities will be given in percentage cumulative monthly.